DASH-01

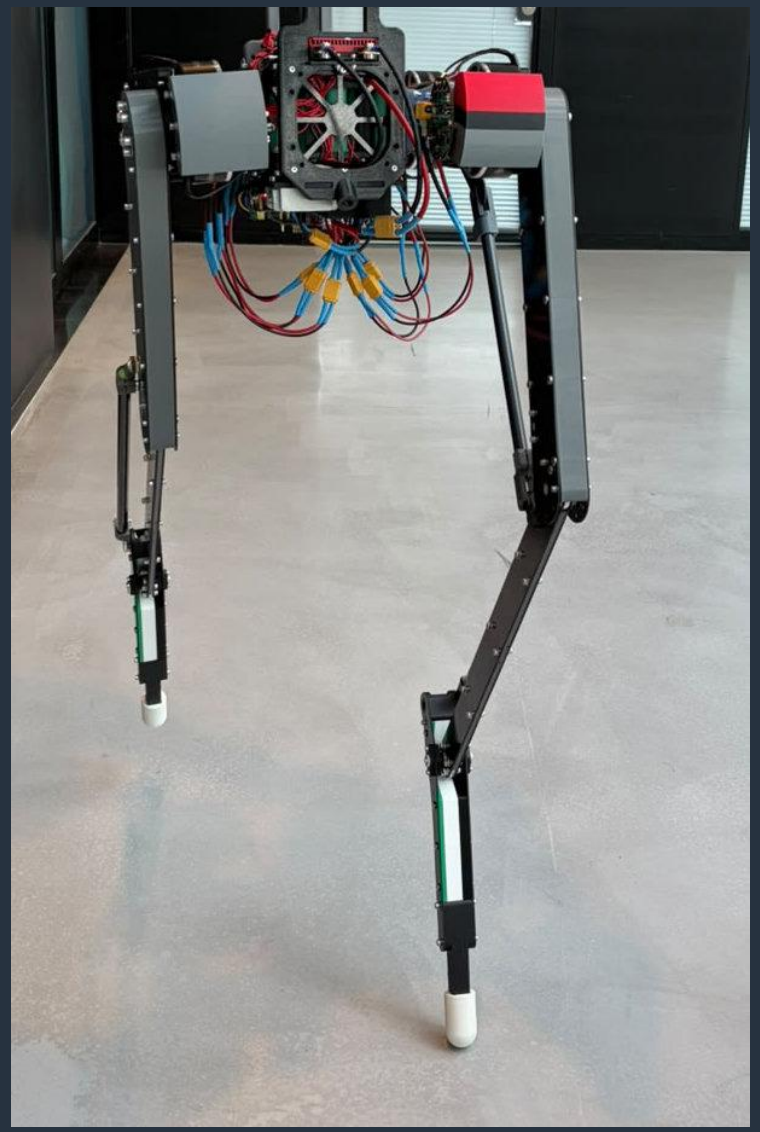
Midterm Presentation · Controller

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Results reviewed 26 October 2026



Previous robot iteration - photo from Tim

Outline

1 **Context**
Existing controller · Semester goal · Starting point

2 **What was done**
Controller · Learning · Gait · Results

3 **Objectives and timeline**
Demo sequence · Early treadmill tests

4 **Challenges**
Sim-to-real gap · Unknown unknowns

PART 1

Context

Existing controller · Semester goal · Starting point

1. Context

2. What was done

3. Objectives and timeline

4. Challenges

Semester goal

INHERITED

Working baseline

Nathann’s RL pipeline
Gait generator + feedback
Deployment code and UI

WHAT CHANGED

A new model

New geometry and flat feet
Corrected mass and inertia
New homing configuration

SEMESTER OBJECTIVE

A short robot demo

Start upright
Walk at commanded speeds
Stop and remain standing

My focus: adapt and validate the learned controller. Tim’s focus: the mechanics.

Starting point: Nathann’s controller

Commanded speed

Sensors
IMU + joints
180 ms history

→

State estimate
Learned velocity
From sensor history

→

RL policy
384 obs + 3 est.
47 outputs · 100 Hz

→

Gait generator
41 gait parameters
6 joint corrections

→

PD + limits
Targets + gains
Torque/speed caps

↓

Six motors + robot

Training only: PPO + privileged critic in MuJoCo / MJX
Ground-truth velocity and contacts help learning, not the deployed actor.

Feedback: joint measurements and IMU → next policy update. Simulation advances in 1 ms substeps.

The policy generates motion targets; motor feedback turns those targets into torque.

PART 2

What was done

Model · Learning · Gait · Results

1. Context
2. What was done
3. Objectives and timeline
4. Challenges

What we explored and what we learned

1 **Stopping on the previous robot** — Older model
Abrupt braking exposed a missing standing skill. Imitation improved quiet holding.

2 **New CAD and gait discovery** — Model + diagnosis
Correct inertials and closed loops; investigate dragging, asymmetry and micro-steps.

3 **Reproduce the walking recipe** — WALK-01
Recover sustained commanded locomotion on CAD-v2 from the supplied homing pose.

4 **Measure real heel/toe support** — Flat-feet baseline
Add contact-based slip, completed-cycle symmetry and mid-stance flat support.

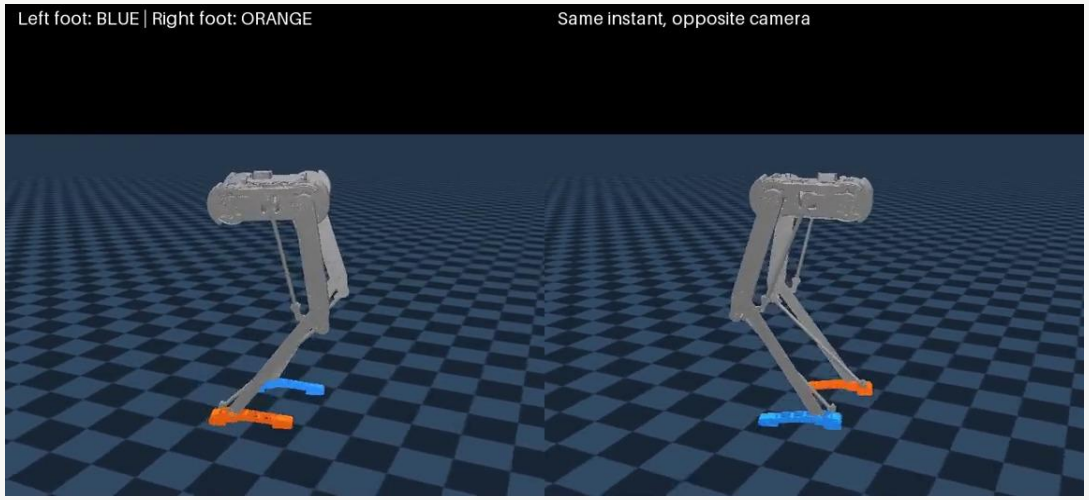
5 **Shape the swing trajectory** — CLEARANCE-01 / 02
Measure clearance; teach higher lift, then target a smaller height band.

Survival, speed tracking and gait quality are different objectives.

Walking policies: visual evolution

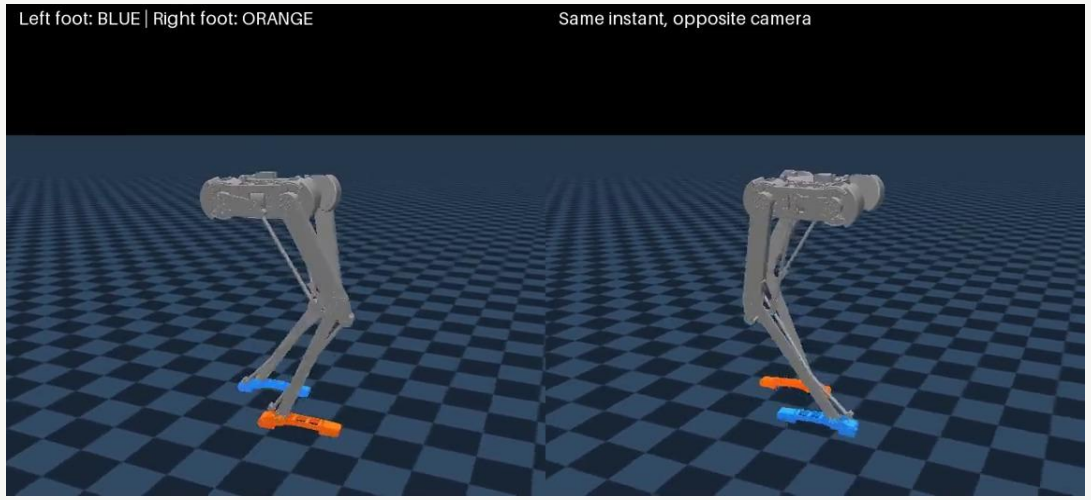
Walking reproduction

WALK-01 best · checkpoint 169.57M



Flat-feet baseline

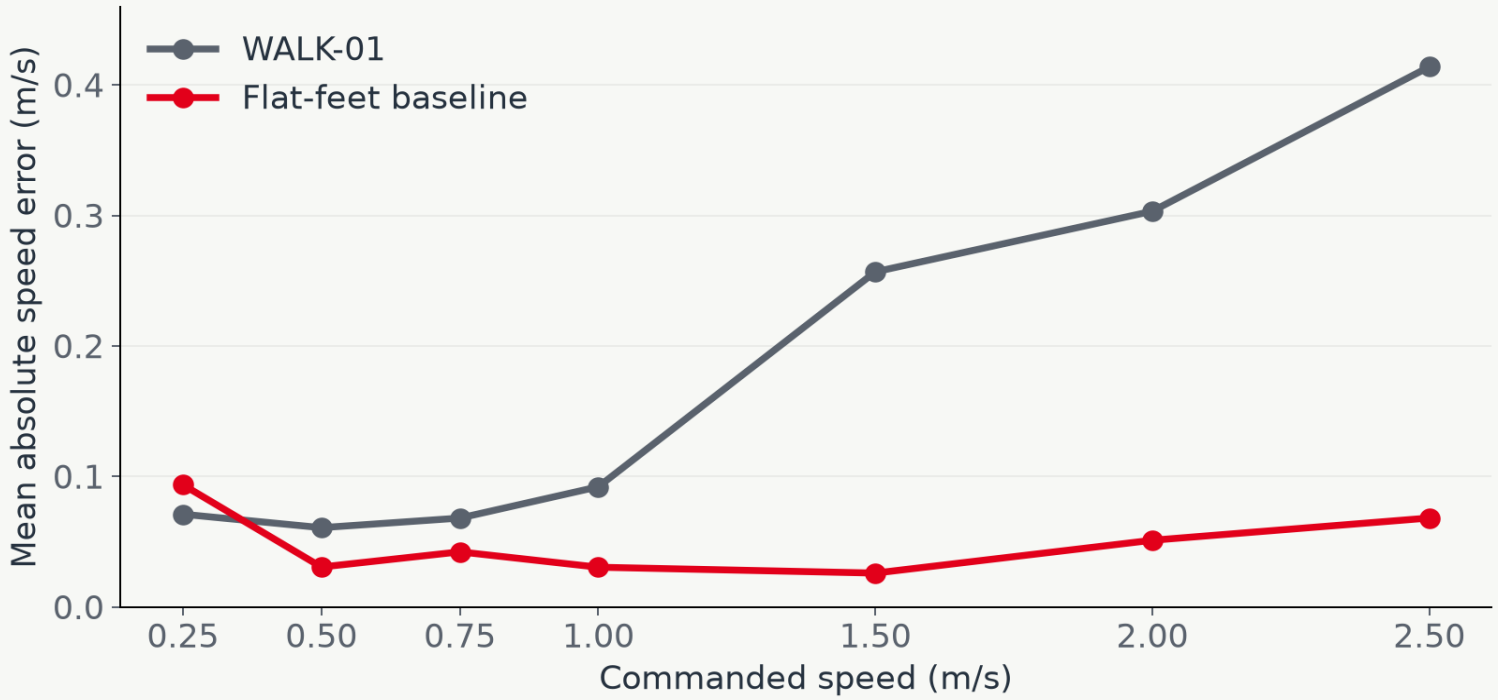
Best checkpoint 184.32M · later used for fine-tuning



Simulation · nominal MJX replays · 12 s excerpts (t = 6–18 s) · 1× playback

Same 1.50 m/s command; different trained policies. Click each clip to play.

Speed tracking



Mean speed error

0.181 → 0.049

m/s across the speed ladder

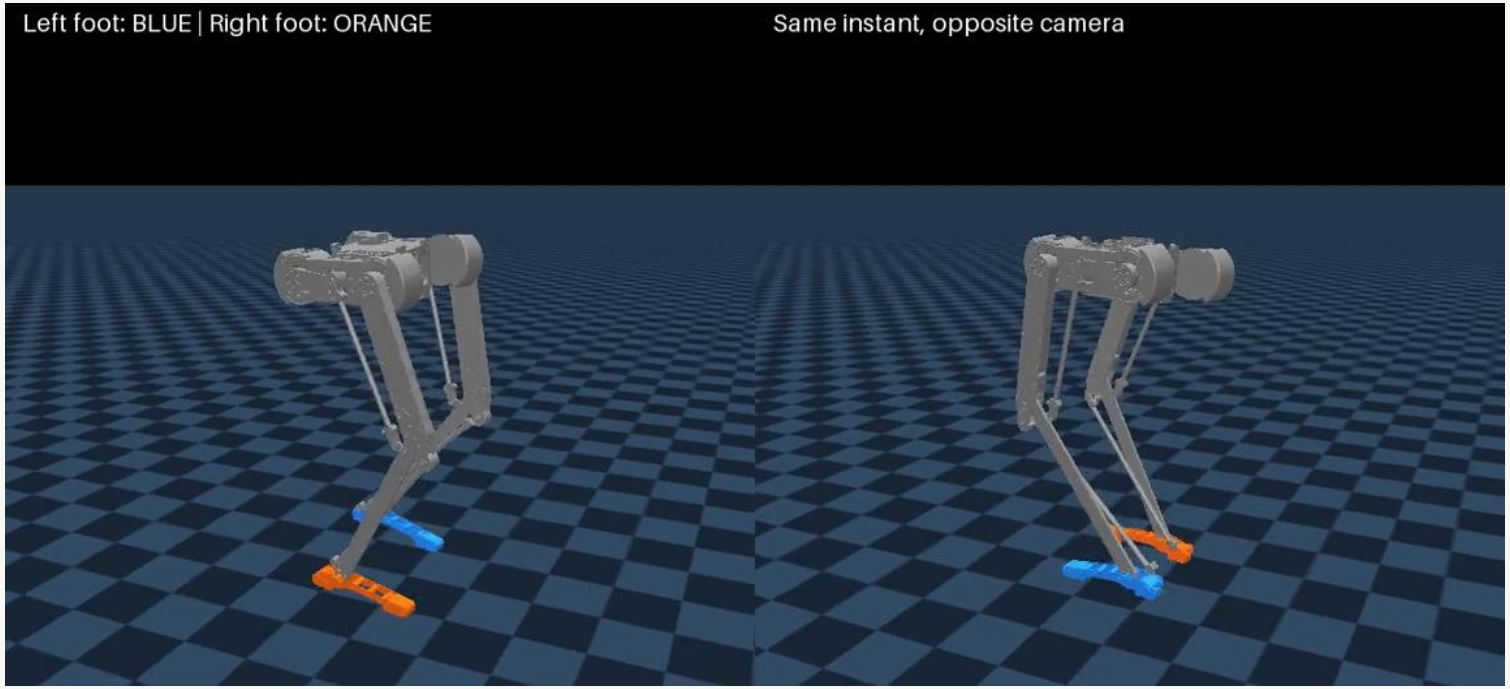
28 / 28

full 30 s trials

both best policies

A better tracking score does not automatically mean a clean or transferable gait.

Foot clearance: first results



Evaluation-protocol replay · trial 0 / 12 · t = 6–18 s · 1×

Mean airborne lift

23.6 mm

left (baseline: 2.46 mm)

20.9 mm

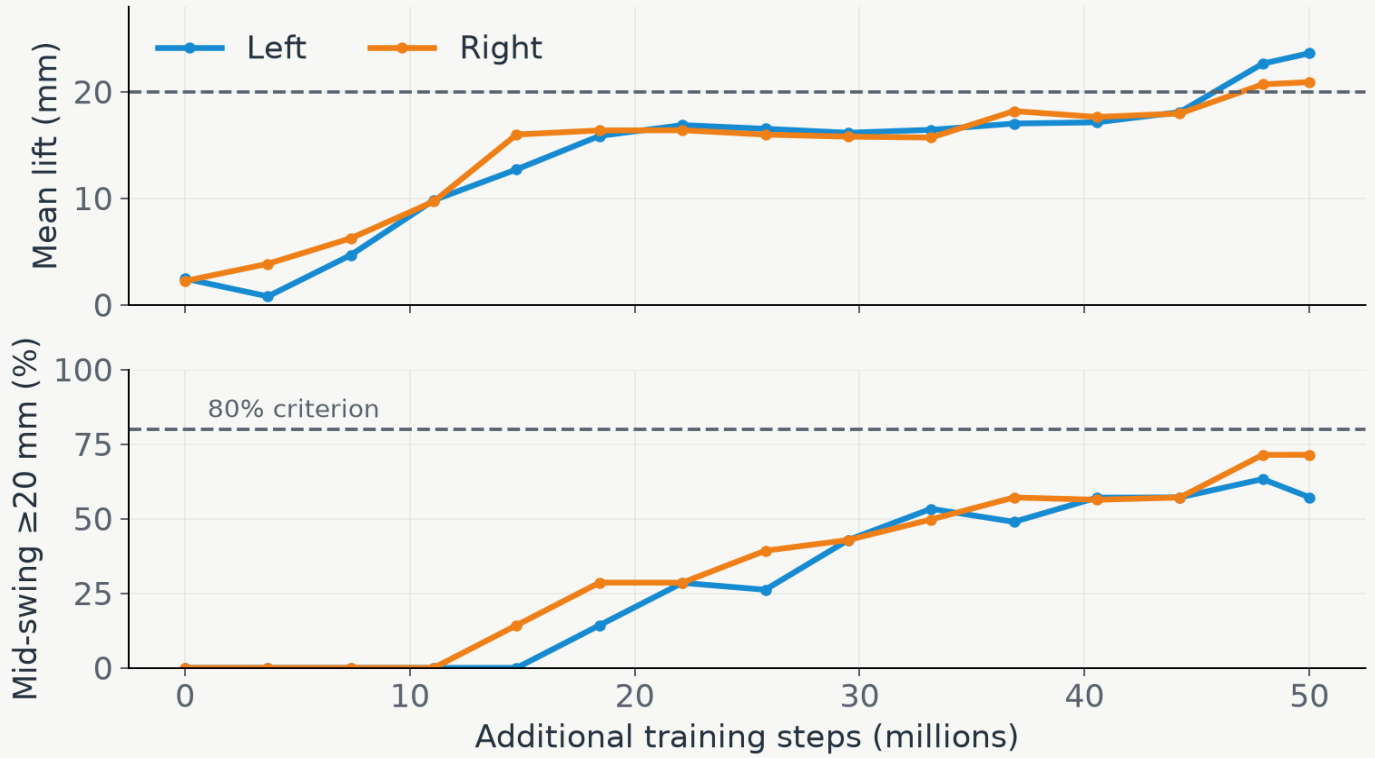
right (baseline: 2.25 mm)

+50M training steps

Target: 1.50 m/s

Promising swing motion in simulation; startup reliability still needs investigation.

Next training objective



NEXT TRAINING

10–25 mm

Minimum swing target

Capped bonus

Penalty above 25 mm

200M steps

Submitted 26 October

Results not reviewed yet

CLEARANCE-02 starts again from the flat-feet baseline, not the high-lift policy.

PART 3

Objectives and timeline

Demo sequence · Early treadmill tests

Objectives for Robotics Day

1 / INITIALIZE

Start upright

Known initial pose

Automatic upright startup

Recover from small disturbances

2 / START

Walk on command

START command

Joystick-style speed changes

Stay within walking regime

3 / STOP

Brake and stand

Smooth deceleration

Stop without assistance

Remain still for ≥5 seconds

These are integration targets — not capabilities already validated on CAD-v2 hardware.

Timeline

NOW / NEXT TEST SLOT	BEFORE ROBOTICS DAY	ROBOTICS DAY	AFTER THE DEMO
SIMULATION			
Compare clearance runs Freeze baseline	Fit measured mismatches Train demo transitions	Freeze tested checkpoint No last-minute retraining	Quantify robustness Write final report
HARDWARE			
Treadmill tests ASAP Calibrate; short trials	Repeat low-speed trials Log and fix failures	Rehearse demo Keep tested fallback	Wider speed envelope Repeatability tests

Repeat throughout: measure → reproduce in simulation → update → retest

First tests: next available slot after assembly, calibration, support and kill checks.

PART 4

Challenges

Sim-to-real gap · Unknown unknowns

Where we are stuck

KNOWN SUSPECTS

The physical model

Friction, compliance, inertia and actuator response

CONTROL AND OBSERVABILITY

Timing and sensing

Delay, velocity estimation and phase timing

REWARD AND VALIDATION

A high score can hide a bad gait

Dragging, contact chatter and fragile startup

UNKNOWN UNKNOWNS

Unmodeled effects

Short trials and logs before adding complexity

We do not know what we do not know until we test the real robot.

What we need

WHAT WE HAVE

- A working RL pipeline
- Better speed tracking
- Higher foot clearance

WHAT COMES NEXT

- Treadmill tests ASAP
- Measure the mismatch
- Integrate start → walk → stop

WHAT WILL HELP

- Test access with Tim
- Repeatable test setup
- Agree demo priorities

Priority: test the real robot while refining simulation.

Questions?

Thank you

Next: treadmill tests and sim-to-real